

10.1 Implementation of Spectral Subtraction for Speech Noise Reduction

Code:

```
clc;
clear;
close all;

Fs = 8000;
t = 0:1/Fs:3;

% Clean signal
x = sin(2*pi*200*t) + 0.5*sin(2*pi*400*t);

% Noisy signal
xn = x + 0.3*randn(size(x));

% FFT of noisy signal
Y = fft(xn);

% Estimate noise spectrum
N = fft(0.3*randn(size(x)));

% Spectral subtraction
xe = real(ifft(max(abs(Y)-abs(N),0).*exp(1j*angle(Y))));
```

```
%% Signals
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot(t,x);
```

```
title('Clean Speech');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
grid on;
```

```
subplot(3,1,2);
```

```
plot(t,xn);
```

```
title('Noisy Speech');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
grid on;
```

```
subplot(3,1,3);
```

```
plot(t,xe);
```

```
title('Enhanced Speech');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
grid on;
```

```
%% Spectrograms
```

```
figure;
```

```
subplot(3,1,1);
```

```
spectrogram(x);
```

```
title('Clean Spectrogram');
```

```
subplot(3,1,2);
```

```
spectrogram(xn);
```

```
title('Noisy Spectrogram');
```

```
subplot(3,1,3);
```

```
spectrogram(xe);
```

```
title('Enhanced Spectrogram');
```

```
%% Magnitude Spectra
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot(abs(fft(x)));
```

```
title('Clean Spectrum');
```

```
xlabel('Frequency Bin');
```

```
ylabel('Magnitude');
```

```
grid on;
```

```
subplot(3,1,2);
```

```
plot(abs(fft(xn)));  
title('Noisy Spectrum');  
xlabel('Frequency Bin');  
ylabel('Magnitude');  
grid on;
```

```
subplot(3,1,3);  
plot(abs(fft(xe)));  
title('Enhanced Spectrum');  
xlabel('Frequency Bin');  
ylabel('Magnitude');  
grid on;
```

Output:



